Mass Spectrometry Data analysis via Graph Machine Learning

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Problem formulation

The mass spectrometry analysis includes two approach: **targeted** analysis and **untargeted** analysis.

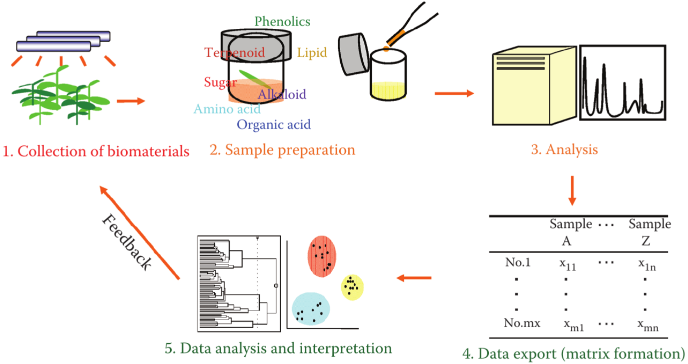


Figure 1: Metabolomics workflow.

Source: [1]

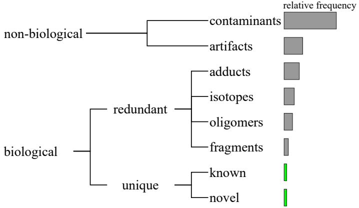


Figure 2: Untargeted composition.

Source: Adapted from [2]

Our objectives

Filter mass spectrometry data via graph machine learning models.

Graph Neural Network (GNN)

It is a function $f : (A, X) \rightarrow Z$ that maps each node $v \in V$ of a graph $G = (V, E)$ to a low-dimensional embedding $z_v \in \mathbb{R}^d$ $(d \ll |V|)$.

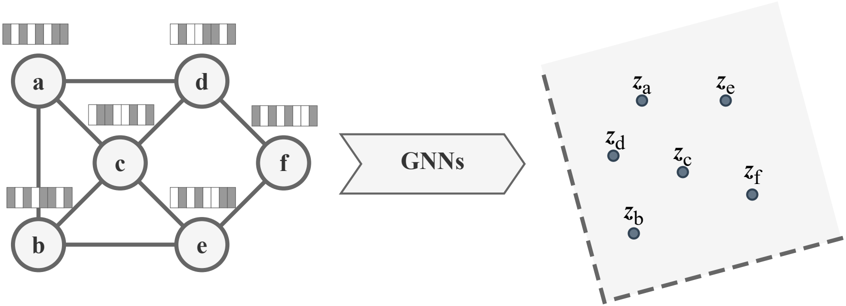


Figure 3: GNNs scheme.

Base GNN models

- VGAE: Variational graph auto-encoders [3]
- ARGVA: Adversarially regularized graph autoencoder for graph embedding [4]
- DGI: Deep graph infomax [5]
- LVGAE Simple and effective graph autoencoders with one-hop linear models [6]
- T-GAE: Transferable Graph Autoencoder for Network Alignment [7].

Mass spectrometry data

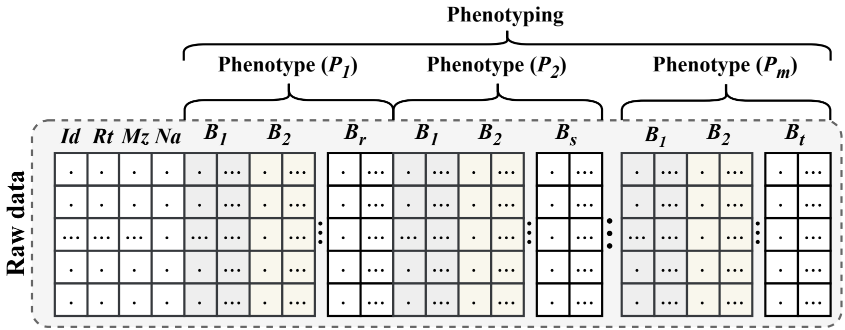


Figure 4: Dataset format.

Approach 1 (matabolomics changes)

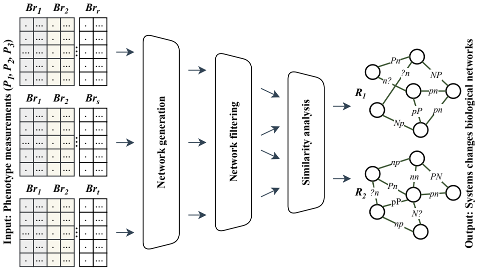


Figure 5: Our workflow.

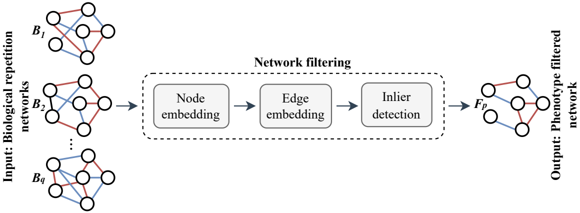
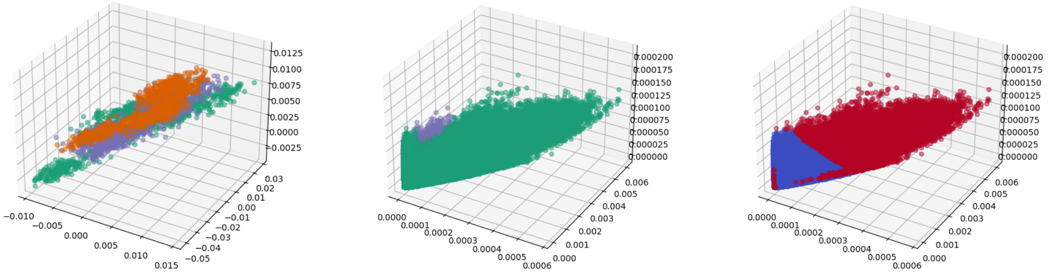


Figure 6: Network filtering.

Approach 1



(a) Node embeddings. (b) Edge embeddings. (c) Outliers.

Figure 7: Embeddings manipulation for *Warming-treated* phenotype on the Leaf.

Approach 2 (network alignment)

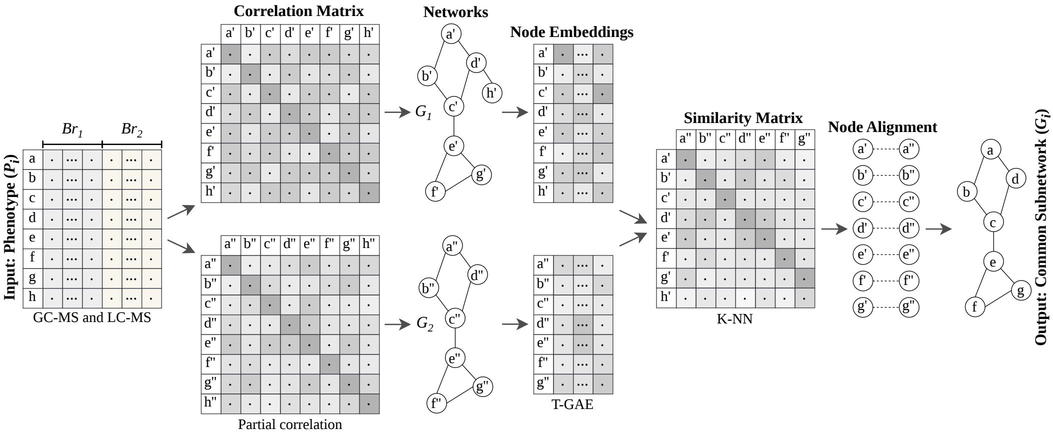
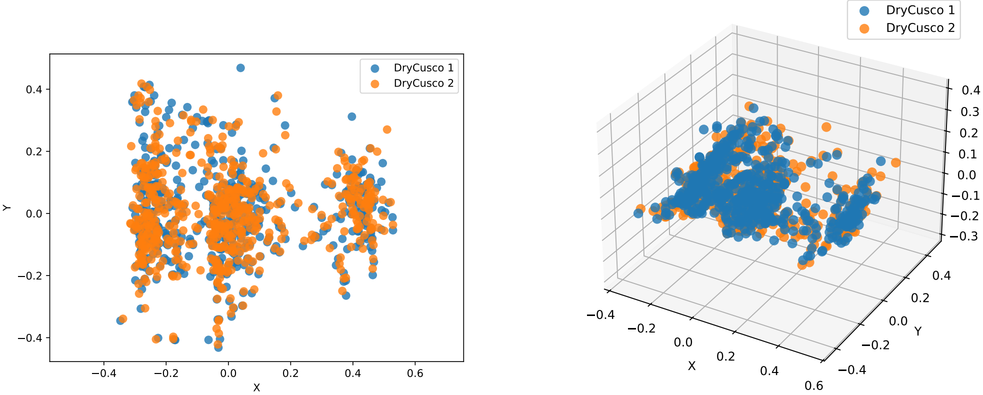


Figure 8: Our workflow.

Approach 2



(a) For Cocoa (DryCusco). (b) For Cocoa (DryCusco).

Figure 9: Embeddings manipulation.

Datasets

These 5 datasets were used in the experiments.

- Mutant [8]
- Leaf [9]
- Mentos [10]
- Cacao (from ICOBA-PUCP)
- Synthetic

Approach 1

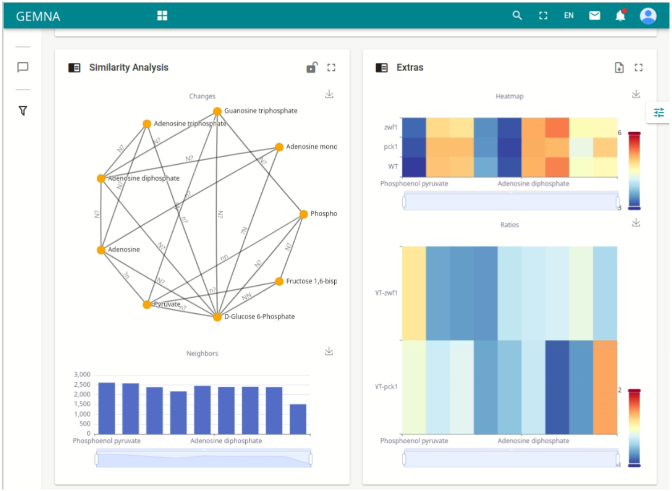


Figure 10: Visualization of analysis MS data.

Approach 2

Table 1: Node aligment (DryCusco).

ID 1	ID 2
0	93
2	2
4	57
5	5
7	93
...	...
542	542
551	551
556	509
557	550
558	530

Table 2: Node filtered **294/559** (DryCusco).

ID	Rt	Mz	Metabolite name
2	0.097	172.95681	Unknown
5	0.201	144.98254	Unknown
15	1.871	118.08675	Betaine
30	1.949	446.88860	Unknown
46	2.110	118.08671	Betaine
531	29.792	374.30350	Docosahexaenoic acid ethyl ester
532	29.794	352.32167	Unknown
537	29.838	454.42642	N-Hexacosanoylglycine
538	29.839	150.12816	2,6-Diethylaniline
542	29.860	398.23344	Drotaverine
551	29.954	150.12807	2,6-Diethylaniline

What is next?

- Perform exhaustive experiments.
- Model metabolomic networks as signed networks.
- Develop a web application.

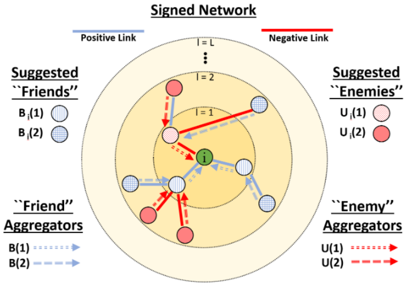


Figure 11: Signed graph convolutional network (SGCN) [11].

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Thanks!

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https://github.com/win7